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Pesticide reduction, climate change, and food security: evidence from rice production in China

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Introduction: The scientific pesticide reduction is important for sustainable agriculture. However, the general public and scholars have been concerned about whether extreme pesticide reduction will lead to a decline in grain yield and thus affect the stability of food security in China.

Methods: This study attempted to evaluate the dynamic impact of pesticide reduction ratio on rice yield using a threshold regression model based on survey data from 1,120 rice farmers in China.

Results: The results show that there is a single threshold effect of the pesticide reduction ratio on rice yield. The threshold value of pesticide reduction of the sample farmers is 33.25%. It suggests that moderate pesticide reduction does not affect rice yield. However, excessive reduction of pesticides is closely linked to diminished pest control effectiveness, which may increase the likelihood of rice yield losses. Furthermore, it can be found that rice pest outbreaks are larger and more frequent under climates such as drought, flooding and high temperature, which could be faced with greater rice yield losses if pesticide use continues to be reduced.

Discussion: The policy makers should not be blind and adventurous, but develop scientific and reasonable pesticide reduction plans according to local conditions. Importantly, dynamic and flexible performance assessment targets for pesticide reduction should be set to cope with the impacts brought by extreme climate change.

KEYWORDS

pesticide reduction, food security, rice yield, pest management, climate change

1 Introduction

The use of pesticides has undoubtedly made a significant contribution to the improvement of pest management and agricultural productivity (Tambet and Stopnitzky, 2021; DuPuis and Christian, 2023). According to the FAO, the active use of pesticides, primarily in the world's major grain-producing and consuming nations during the first 15 years of the 21st century, reduced the probability of yield loss of grain by almost 40%, with a market economic value of 100 billion dollars. More importantly, the use of pesticides has played an important role in alleviating the problem of food security in China, which has a large population (Hou et al., 2020; Abebe and Amare, 2025). However, the long-term overuse of pesticide has also brought about a series of negative consequences for international trade, ecosystems, and human health (Chatzimichael et al., 2020; Balakumar and Radhakrishnan, 2025).

To promote the green transformation of traditional agriculture, the Chinese government has prioritized the implementation of a sustainable development strategy with a strong focus on environmental conservation, and implemented a "zero-growth" pesticide use program in 2015 (Pan et al., 2021; Tambo et al., 2024). By 2023, the cumulative reduction of total pesticide use in China

amounted to 34.14%, with some provinces experiencing rapid pesticide reduction. For example, Gansu Province in China has reduced the amount of pesticide by 45%, and Hainan Province has reduced the amount of pesticide by 50%. During this period, some scholars even benchmarked countries such as Europe and the US, believing that China's pesticide reduction potential of more than 60% (Cai et al., 2021).

However, China's pesticide reduction efforts have advanced significantly. At the same time, the public and scholars are deeply concerned about whether the ongoing reduction in pesticide use will impact China's food security stability (Yang et al., 2018; Washuck et al., 2022; Schneider et al., 2023). This is mainly because biopesticide technology has not yet fully matured and the adoption rate of IPM strategies remains low (Wossink, 2007; Yan et al., 2024; Victorel et al., 2024). For example, some early studies found that crops face a greater risk of reduced yields when severely affected by pests and diseases (Zhao et al., 2024; Abdel-Dayem et al., 2025). While technologies such as integrated pest management (IPM), sustainable intensification, or ecological pest control are more environmentally friendly, their effectiveness and efficiency in suppressing pests in the short term remain insufficient for field crops (Huang et al., 2022; Zanzana et al., 2024; Rao et al., 2025). However, rampant pest infestations are one of the key factors causing fluctuations in crop yields and threatening the stability of regional food supply. Pesticides remain an important and effective means of managing pests in food crops (Xie et al., 2019), which is crucial for stabilizing grain production and guaranteeing the supply of food security in China (Huang et al., 2024). Moreover, in recent years, some agricultural areas in China have experienced frequent climate changes such as drought, floods, high temperatures, and frost. These extreme weather conditions not only induce high frequency and widespread occurrence of crop pests, increase the difficulty of pesticide reduction, but also cause reduced grain production and crop failure, posing risks to China's food security stability.

The results of scholars' discussion on whether pesticide reduction will lead to a decrease in grain yield are inconclusive (Pan et al., 2021; Cai et al., 2021). Some scholars argue that pesticide reduction lead to an increased risk of grain yield losses (Zhang et al., 2022; Na et al., 2024), because pesticides are essential inputs of agricultural production input. Yan et al. (2020) found a positive correlation between pesticide intensity and grain yield. In addition, the results of field experiments conducted by Wu et al. (2021), that pesticide reduction can directly lead to rice yield losses in the absence of other remedial measures. Other scholars argue that pesticide reduction will not affect grain yield and may even have the potential to boost grain yield (Mohring and Finger, 2022). For example, by reducing the frequency of pesticide use by 25% or reducing the dose of pesticide use by 30% in some field experiments, no significant changes in rice panicle number, 1,000-grain weight, and yield were found (Castilla et al., 2020; Wies et al., 2022). And it has also been shown that pesticide reduction facilitates the reproduction of natural enemies of crop pests such as bees and insects, which enhances rice yield through their promotion of rice pollination and root tillering (Rdiger et al., 2024). It is evident that discussions regarding whether reducing pesticide use will affect grain yields must focus on specific environmental conditions.

The inconsistency in existing research conclusions may be due to two possible reasons: Firstly, scholars mostly set the correlation between pesticide reduction and grain yield as a linear relationship by default, and have not yet accurately revealed the impact mechanism and stage evolution characteristics of pesticide reduction on grain yield. Because pesticide is typically a damage control input factor compared to other agricultural input elements such as seed, fertilizer, and machinery (Qin and Lv, 2020; Rashidi et al., 2024). The main goal of pesticide use in agriculture is to prevent pest outbreaks. However, farmers are mostly risk averse and they tend to exceed scientific dosage recommendations and come to use pesticide at high frequency or in large doses in practice (Benoît et al., 2020). The reduction of excessive pesticide use will obviously not affect grain yield, indicating that the correlation between pesticide reduction and grain yield is likely to be non-linear. Secondly, scholars have different sample reference targets and scenarios when discussing the potential for pesticide reduction (Yang et al., 2018), resulting in significant differences in the conclusions of pesticide reduction potential assessment and a lack of practical guidance value. Extreme weather conditions, such as droughts, floods, high temperatures, and frost, tend to exacerbate the occurrence of crop pests (Washuck et al., 2022). Consequently, farmers are compelled to increase pesticide usage as part of scientific pest management in practices. Under such circumstances, the potential for pesticide reduction is significantly diminished.

However, scientifically assessing the influence of pesticide reduction on grain yield and exploring scientific methods of pesticide reduction in the context of food security are important in realizing sustainable agricultural development. This work is an important practical guidelines for next stage of sustainable pesticide reduction in China and other countries. Yang et al. (2018) also clearly pointed out that pesticide reduction in China cannot go to the extreme. China cannot implement extreme reductions without considering potential impacts on grain yield (Chatzimichael et al., 2020; Benoît et al., 2020). Unfortunately, with the limitations of macro-statistical data, we are unable to currently observe the precise effects of pesticide reduction on food security stability in China. Therefore, this study will attempt to analyze the mechanism of pesticide reduction on grain yield using data from 1,120 rice farmers' surveys in China.

The contribution of this study is mainly: First, a threshold regression framework was established for empirical validation, evaluating the nonlinear dynamic effect of pesticide reduction ratio on rice yield. It was found that the moderate pesticide reduction in the early stage did not affect rice yield. However, as pesticide reduction continues, excessive reductions below the scientifically recommended dose lead to yield losses. Second, taking into account the effects of realistic extreme climates (drought, flood, high temperature, and frost) on induced pests, this study demonstrates the variation in the impact of pesticide reductions on rice yield under different climatic conditions. Third, this study focused on rice farmers as the primary research subjects, enabling observation of real-world pesticide reduction practices. Because the vast majority of rice farmers in China use the rice they grow as their food ration, they desire food safety and prefer fewer pesticide or even no pesticide. So extensive micro-level data were collected on pesticide reduction or even no pesticide use. The empirical results offer actionable insights for target setting and dynamic adjustment optimization of pesticide reduction in various regions under different scenarios.

2 Materials and methods

2.1 Research hypothesis

2.1.1 Impact of pesticide reduction on rice yield

Previous scholars have assessed the current status of pesticide use by grain growers in China through methods such as the

Cobb–Douglas production function, the Damage Control Function, and the transcendental logarithmic function, all of which have found that the phenomenon of excessive and inefficient use of pesticide is prevalent (Sun et al., 2020). That is, the real dose of pesticide used by farmers exceeds the scientific standard of use for rice pest control (Vasco et al., 2021; Cheng et al., 2025). However, whether pesticide reductions affect rice yield is largely determined by the effectiveness of pest management (Castilla et al., 2020). Generally, only the right amount of pesticide is needed for rice pest management (Sun et al., 2020), too much is a waste, and too little will affect the effectiveness of pest management. Therefore, this study concluded that the sample farmers who used pesticide excessively were bound to go through at least two stages of pesticide reduction:

The first stage is the reduction phase for overdose pesticide, whose reduction does not affect rice yield. For example, pesticide mismatch and pesticide mixing due to the low educational qualities of farmers, and inefficient use of pesticide due to inefficient machinery such as manual sprayers (Tambet and Stopnitzky, 2021). Reducing these irrational pesticide uses by providing agricultural technology training, equipment improvement, and socialization services does not affect the effectiveness of pest control and rice yield (Pan et al., 2021). The second stage is the continued reduction of pesticide below the scientific use dose standard, which may reduce the effectiveness of pest management and result in an increased risk of rice yield loss. This is because pest management during crop production in China is still highly dependent on the use of pesticides. And the use of pesticides in large quantities is still an important means of ensuring stable rice yield (Xie et al., 2019). The pesticide reduction is likely to lead to large outbreaks of pests such as rice planthopper, rice blast, and rice stem borer, which have developed mutations and resistance, in a short period (Benoît et al., 2020). Increased pest and weed damage could result in loss of rice yield or even crop failure.

To understand the above logical relationships from a scientific perspective grounded in mathematical derivation, we construct a Damage Control Function:

$$Y = A \cdot f(X) \cdot G(Z) \tag{1}$$

Equation 1, Y represents rice yield. A denotes Hicks-neutral technological progress. $f(X)$ represents the standard production function. X represents conventional production factors (such as land, labor, and fertilizer). $G(Z)$ is the damage control function. Z is the damage control input (pesticide), typically ranging from [0, 1]. This indicates that pesticides can control potential losses within a certain proportion. $G(Z)$ is typically defined such that $G(Z) \geq 0$, $G(Z)' > 0$, and $G(Z)'' < 0$, meaning the marginal damage control effect of pesticide application is positive and diminishing. Next, taking the derivative of rice yield Y with respect to pesticide input Z:

$$\frac{\partial Y}{\partial Z} = A \cdot f(X) \cdot G(Z)' \tag{2}$$

Since $A > 0$, $f(X) > 0$, and by the properties of the damage control function $G(Z)' > 0$, it follows that $\frac{\partial Y}{\partial Z} > 0$ in Equation 2. This implies that, holding other conditions constant, increasing pesticide input (Z) can significantly reduce yield losses caused by pests by enhancing damage control efficiency $G(Z)$, thereby exerting a positive impact on rice yield per unit area. However, since $G(Z)'' < 0$, this positive impact exhibits diminishing marginal returns. That is, pesticide inputs substantial benefits initially, but beyond a certain threshold, their contribution to yield increases gradually diminishes. Therefore, we can deduce through reverse reasoning that the reduction in pesticide usage exerts a nonlinear effect on rice yields.

Based on the comprehensive analysis of the theoretical framework, this investigation postulates the following hypothesis (H1): There is a threshold effect of pesticide reduction on rice yield. That is, moderate pesticide reduction does not affect rice yield, but excessive pesticide reduction leads to an increase in pests, resulting in an increased probability of rice yield loss.

2.1.2 Impact of extreme climate on pesticide reduction and pest occurrence

The occurrence of pests will show a certain cyclic pattern with the growth stages of rice panicle initiation, booting, heading, anthesis, milky ripe, etc. However, the cyclical patterns of pest outbreaks are easily disrupted by extreme weather events such as droughts, floods, heatwaves, and frosts (Garcia et al., 2020; Nafeesa et al., 2024). This can also result in further deepening of pest outbreaks, which in turn increases the difficulty of pesticide reduction under the goal of maintaining food security (Haq et al., 2023). Overall, the impacts of extreme climate are reflected in three main aspects:

Firstly, extreme climate increases crop susceptibility to pests. Rice growth often requires moderate conditions of light, heat, and moisture, so extreme climates can damage the immune system of rice (Washuck et al., 2022). Secondly, some studies have shown that weather such as low temperature and rainfall will also reduce the activity of natural enemies of pests (Nohara, 2024; Rdiger et al., 2024). Extreme climates such as drought and high temperature can inhibit the egg-laying and feeding of beneficial insects. Thirdly, extreme weather may be able to promote the growth and development of pests (Garcia et al., 2020). For example, conditions such as humidity, dryness, and high temperature can stimulate pest evolution and egg-laying, and promote pest migration and aggregation (Benoît et al., 2020). In general, the use of pesticides in extreme climatic conditions is potentially higher compared to normal years, otherwise, there is a risk of large-scale and high-frequency outbreaks of pests.

Similarly, we interpret this logic through mathematical deduction. The severity of "damage" in the Damage Control Function is not fixed. It is strongly influenced by climate. We treat climate as a key exogenous shock affecting the base incidence and severity D of pest outbreaks. In the theoretical model, the climate variable C exacerbates potential rice yield losses, meaning D is an increasing function of C. Furthermore, the climate variable C shapes the form of the function $G(Z)$. Specifically, during extreme climate events (increased C), conditions for pest survival and spread improve, elevating both the risk and scale of outbreaks. This implies that at any given pesticide input Z, the "damage baseline" requiring control expands. This can be interpreted as an increase in climate variable C raising the marginal output of the damage control function $G(Z)$. Therefore, we calculate the cross

partial derivative of rice yield with respect to pesticide input Z and climate C:

$$\frac{\partial^2 Y}{\partial Z \partial C} = A \cdot f(X) \cdot \frac{\partial^2 G}{\partial Z \partial C} \tag{3}$$

If $\frac{\partial^2 G}{\partial Z \partial C} > 0$, then $\frac{\partial^2 Y}{\partial Z \partial C} > 0$ in Equation 3. This indicates that in environments with higher climate risk (C), the marginal utility of pesticide input (Z) for stabilizing and enhancing rice yields (Y) increases. In other words, under extreme climate conditions, pesticide use becomes more critical for ensuring food security.

Therefore, this study empirically formulate the hypothesis (H2): The probability of pest outbreaks increases under extreme climate conditions. That is, pesticide reduction efforts may face greater challenges during extreme weather events while maintaining stable grain yields.

The research framework of this study is shown in Figure 1.

2.2 Data

The dataset used in this study came from a rural household survey conducted by the research group in August 2023 in the provinces of Hubei and Henan, China. The above two provinces are the main grain producing regions in China. For major grain-producing regions, the data on pesticide reduction practices can be more representative. To ensure the sample's representativeness across key socioeconomic and agricultural dimensions, a multi-stage random stratified sampling method was employed. The specific steps were as follows: First, all counties in Hubei and Henan were stratified based on their gross grain output and per capita agricultural GDP to capture regional economic disparities. From each stratum, we randomly selected five counties, resulting in a total of 10 sample counties. Within each selected county, two townships were randomly chosen. Subsequently, within each township, four villages were randomly selected from a list stratified by village population size, ensuring the inclusion of villages with varying scales and production characteristics. Finally, within each village, rice farmers were randomly selected from the official household roster provided by the village committee. This approach aimed to minimize selection bias and ensure that farmers with different operational scales and demographic backgrounds had an equal probability of being included. A total of 1,120 valid questionnaires were completed from 80 villages across 20 townships (Figure 2). The survey questionnaire was completed through face-to-face interviews at farmers' homes. Household heads or major agricultural production decision-making members are selected as interviewees, and the investigator uniformly asks questions and fills out questionnaires. The questionnaire design mainly revolves around the pesticide use, climate, and rice yield of farmers. Our research collected and organized responses from rice farmers. It should be noted that most of our survey data is derived from farmers' recollections and oral accounts. To enhance data quality control, we repeatedly verify any discrepancies in farmers' responses during the survey process, thereby minimizing the potential for misunderstandings and misreporting. We declare that this investigation has been approved by the Academic Ethics Committee of the School of Law and Business, Wuhan Institute of Technology (No. wit2024102809). The collection method of farmer questionnaires were carried out accordance with the Declaration of Helsinki. We confirm that we have obtained informed consent from all respondents. We orally stated the academic purpose, requirements, and confidentiality agreement of the survey during the interviews, and ensured that informed consent was obtained from all respondents.

It should be noted that rice was selected as the research object for two main reasons: Firstly, rice is the highest-yielding grain crop in China, with a wide planting area and significant potential for pesticide reduction (Sun et al., 2020). Secondly, many small-scale farmers in China experience the phenomenon of "both eating and selling" after rice harvest. That is, farmers will divide their rice into two parts, one for daily consumption and the other for sale (Qin and Lv, 2020). Importantly, to ensure the food safety of rice, farmers often reduce or even do not use pesticide (The pesticide reduction ratio is 100%) in their rice fields. This provides a large number of real case samples for the continuity of pesticide reduction data in this study, which is very beneficial for observing the dynamic changes in the impact of continuous pesticide reduction on grain yield.

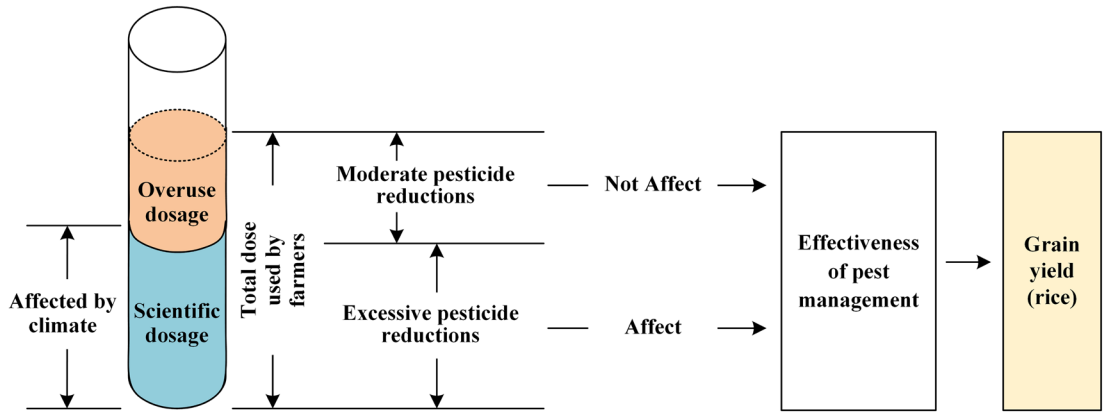


FIGURE 1
Analysis framework of this study.

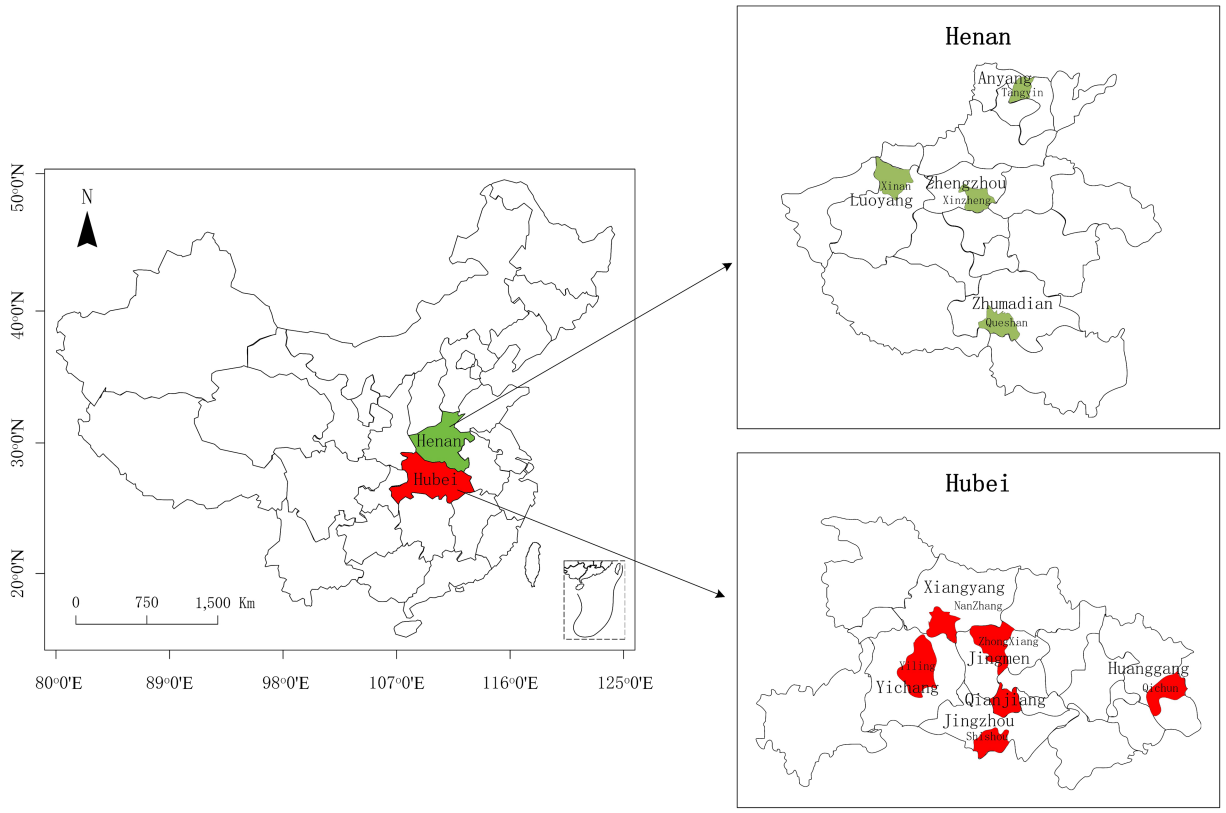


FIGURE 2
Distribution of sample farmers in this study.

2.3 Model

From the theoretical analysis section, it can be seen that due to the non-linear impact of pesticide reduction ratio on rice yield, traditional OLS linear estimation models cannot accurately evaluate the dynamic impact of different pesticide reduction ratio on grain yield. The theoretical analysis indicates that as pesticide reductions increase, the impact of pesticide use on rice yields will reach a turning point. So, this study uses a threshold regression model to estimate the segmented impact of pesticide reduction on grain yield. Assuming that only when the pesticide reduction ratio exceeds a specific value γ, the impact of pesticide reduction on grain yield will change. Therefore, the following segmented functions can be constructed:

$$\begin{cases} yied_i = \mu_1 + \beta_1 reduce_i + \varepsilon_i, if\ reduce_i \le \gamma \\ yied_i = \mu_2 + \beta_2 reduce_i + \varepsilon_i, if\ reduce_i > \gamma \end{cases} \quad (4)$$

where, $yied_i$ represents the rice yield per unit area of the i-th farmer. $reduce_i$ is the pesticide reduction ratio. γ is the threshold value for pesticide reduction to be estimated. μ_1 and μ_2 are intercept terms. ε_i is random interference term. β_1 and β_2 are the impact coefficients to be estimated. Further, the Equation 4 can be merged into:

$$\begin{aligned} yied_i = {} & \mu_0 + \beta_1 reduce_i \cdot \varphi(reduce_i \le \gamma) \\ & + \beta_2 reduce_i \cdot \varphi(reduce_i > \gamma) + \varepsilon_i \end{aligned} \quad (5)$$

where, $\varphi(reduce_i \le \gamma)$ and $\varphi(reduce_i > \gamma)$ are characteristic functions. That is, when the pesticide reduction ratio of the i-th farmer meets the range in parentheses, the characteristic function takes a value of 1, otherwise it takes a value of 0. If the estimated impact coefficient $\beta_1 \neq \beta_2$, it indicates the existence of threshold effect. However, it is difficult to estimate both the threshold value γ and coefficient β_i simultaneously, as this is a parametric nonlinear function. Therefore, we borrowed the method of Hansen (1999), and solved the distribution function of γ by using the sum of residual squared $SSR(\gamma)$ estimated by the model, and constructed the confidence interval of $\hat{\gamma}$ to test the existence of the threshold value γ. Obviously, when the threshold value γ is known, the value of the characteristic function can be determined based on each observation value $reduce_i$, and Equation 5 can be converted into an easily estimated linear function. The heterogeneity of the effects of different pesticide reduction ratio on rice yield can be observed through the significance of the coefficients.

To verify the impact of pesticide reduction ratio on grain yield under extreme climate conditions, we need to examine whether extreme climate plays a regulatory role in this impact. That is, under extreme climate conditions, whether the impact of pesticide reduction on rice yield would change. Currently, there are two methods for adjusting the corresponding validation approach: group regression or interaction terms. Considering that grouping estimation may lead to a loss of effective sample size, we have chosen to construct indicator interaction terms in conventional moderation testing

methods to evaluate the moderating effects of extreme climate. The model settings are as follows:

$$yied_i = \mu_0 + \beta_3 reduce_i + \beta_4 climate_i + \beta_5 reduce_i \times climate_i + \alpha_i control_i + \varepsilon_i \quad (6)$$

where, $climate_i$ refers to the extreme climatic conditions encountered during rice production (drought, floods, high temperatures, and frost). $control_i$ are another control variables that may affect rice yield, including variables such as personal, family, and production characteristics of farmers. The impact of extreme climate can be comprehensively judged based on the estimated coefficients β_3, β_4, and β_5.

Dependent variable: Rice yield. Rice yield serves as a proxy indicator for measuring the degree of threat to food production stability in this study. Higher yield losses caused by pests indicate weaker resilience of the local food production system against pest stress, thereby posing a greater threat to its stability. The average yield of paddy per mu harvested by the sample rice farmers in 2021 is utilized in this study to measure this index.

Independent variable: Pesticide reduction ratio. Currently in the research literature related to pesticide reduction, pesticide use in farmers' production is mostly measured by indicators such as pesticide use intensity per unit area, pesticide use frequency, or pesticide use cost (Bekken et al., 2022). This study focused on the ratio of reduction in pesticide use intensity per unit area of rice pest management. Therefore, the percentage reduction in the pesticide dose used by rice farmers in the current year compared to the pesticide dose used in the previous year was used to measure the pesticide reduction ratio. To validate the reliability and consistency of our findings, we also utilized the absolute amount of pesticide reduction and pesticide intensity as proxy indicators for estimation.

Moderating variable: Extreme climate. Extreme climate can disrupt ecological balance and greatly alter the growth and breeding environment of crops, pests, and natural enemy organisms. Rice has water-loving and warmth-loving characteristics, so the negative effects of climate extremes such as drought, flood, high temperature, and frost are greater (Garcia et al., 2020). Therefore, the heterogeneous effects of the four extreme climates, drought, flood, high temperature, and frost, were sequentially tested in the model by setting dummy variables.

Control variables: These mainly refer to other variables that may affect rice yield, including input factors (pesticide, fertilizer, machinery), personal characteristics of farmer (gender, age, education), family characteristics (income, agricultural labor), production characteristics (rice scale, variety, and irrigation condition), and regional dummy variables (Guo et al., 2022; Bashir et al., 2025). Among them, input factors are the core variables that affect grain yield per unit area. Farmers' personal and family characteristics affect decisions on rice production, inputs, and management. Production characteristics affect rice production efficiency and patterns (Zhuang et al., 2022). Of course, considering the differences in climate, soil, and growth cycle of rice production at the geographic region level, the province is used as a dummy control variable in the model. The variables are shown in Table 1.

3 Results

3.1 Threshold regression estimation results

The necessary covariance diagnosis was performed for each variable in the model. Next, the pesticide reduction ratio of the sample rice farmers was set as a threshold variable, and the Stata software self-help method and likelihood ratio test were utilized to verify the

TABLE 1 Definition and descriptive statistical analysis of variables in the model.

Variables	Definition and assignment	Mean	SD
Rice yield	The average yield of rice per unit area in 2021 (kg/mu)	606.25	26.72
Pesticide reduction ratio	Pesticide reduction ratio per unit area for rice pest management (%)	20.37	3.69
Drought	Whether drought occurs during the rice growth cycle: Yes = 1, No = 0	0.28	0.12
Flood	Whether flood occurs during the rice growth cycle: Yes = 1, No = 0	0.17	0.02
High temperature	Whether rice undergoes high temperature during growth cycle: Yes = 1, No = 0	0.35	0.10
Frost	Whether frost occur during the rice growth cycle: Yes = 1, No = 0	0.19	0.05
Pesticide intensity	The amount of pesticide used per unit cultivated area of rice (kg/mu)	0.26	0.19
Fertilizer intensity	The amount of fertilizer used per unit cultivated area of rice (kg/mu)	85.72	16.23
Mechanical input	Unit cost of mechanical input during the rice growth cycle (yuan/mu)	315.25	82.24
Gender	The gender of the interviewee: Male = 1, Female = 0	0.46	0.12
Age	The age of the interviewee (year)	48.50	7.63
Education	The education level of the interviewee (year)	7.31	1.26
Family income	The total income of family members (10,000 yuan)	9.80	2.17
Agricultural labor	Number of agricultural labor in family members (person)	2.07	0.58
Rice scale	The scale of rice planting in 2021 (mu)	12.09	5.47
Irrigation condition	The completeness of irrigation in the field: Poor = 1, General = 2, Good = 3	2.58	0.64
Variety	Whether the rice planting mode is double cropping: Yes = 1, No = 0	0.69	0.10
Henan	Whether the sample belongs to Henan Province: Yes = 1, No = 0	0.31	0.25

15 mu = 1 ha in China.

existence of the threshold value. The results in Table 2 were obtained by performing the single-threshold and double-threshold tests sequentially using repeated sampling. It showed that the pesticide reduction ratio variable passed the single threshold test and its *F*-value passed the 5% significance level test and the threshold value $\gamma = 33.250$ was obtained. However, the double threshold test result of the model is no longer significant, indicating the existence of a single threshold for the pesticide reduction ratio. Therefore, further empirical analysis can be done using threshold regression estimation.

The results of further estimating the effect of pesticide reduction on rice yield using threshold regression are shown in Table 3. Based on the results of M1 obtained from the baseline regression using the core explanatory variables, the estimation results of M2 were obtained by incorporating the control variables. It showed that the coefficient of effect of pesticide reduction ratio on rice yield did not pass the test of significance when the pesticide reduction ratio was less than 33.250%. However, the coefficient of the negative effect of pesticide reduction ratio on rice yield passed the significance level test of 1% when the pesticide reduction ratio is higher than 33.250%. This shows that pesticide reduction leads to a significant increase in the probability of rice yield loss when the rate of persistent pesticide reduction is greater than 33.250%. The research hypothesis H1 is verified. Therefore, scientific pesticide use remains a key element in ensuring grain productivity. Pesticide reductions should not be carried out blindly and aggressively during rice cultivation. Of course, the results of the samples also reflect that modest pesticide reductions do not result in rice yield losses, and are much less likely to jeopardize the safe and stable supply of grain in China. Regulating pesticide use by rice farmers through social services, green control technologies, and organizational changes can reduce excess pesticide use without yield loss.

The identified pesticide reduction threshold of 33.25% is not merely a statistical cutoff point but carries profound biological and economic implications. Biologically, this threshold represents the buffer capacity boundary of a farmland ecosystem's natural pest control capabilities—such as predator regulation and crop resistance—and aligns with the "control index" theory in integrated pest management. When reduction rates fall below this threshold, residual pesticide inputs and ecosystem services collectively suppress pest populations below economically damaging levels, ensuring stable yields. Exceeding this threshold, however, breaches natural control capacities, sharply increasing outbreak risks and significantly raising the probability of yield losses. Economically, this threshold precisely delineates the equilibrium point between marginal benefits and marginal costs of pesticide reduction. "Moderate reduction" (below 33.25%) constitutes a Pareto improvement, yielding environmental and health benefits (positive marginal benefits). Conversely, "excessive reduction" (above 33.25%) entails sharply increased marginal yield losses, where costs far outweigh the benefits gained from reduced pesticide use. Thus, this threshold provides critical quantitative evidence for formulating scientifically grounded reduction policies that are "neither reckless nor overly cautious," delineating a feasible range that achieves a win-win outcome for both yield security and ecological safety.

The coefficients of the effect of extreme climate on rice yield all passed the significance test, and the extreme climate significantly increased the probability of rice yield loss. Specifically, the negative effect of drought on rice yield passed the 1% significance level test. Since rice is a water-loving crop, persistent drought will cause water evaporation and soil consolidation in paddy fields, which in turn will lead to the drying up of plant roots and stems, inability to absorb fertilizers, and shortening of paddy maturity. The negative effect of flood on rice yield was passed the 5% level of significance test. The occurrence of flooding affects the normal drainage and irrigation management of rice fields. Flood of rice fields impairs the respiration of the rice root system and causes rice plants to collapse, leading to an increased probability of rice yield loss. The negative effect of high temperature and frost on rice yield passed the significance level test of 5% and 10%, respectively. Natural science experiments have shown that suitable temperature conditions are required throughout the growth of rice. Sustained high temperature can cause a series of negative effects on rice such as tillering disruption, pollen inactivation, and reduced fruiting rate. Frost weakens the physiological activity, respiration, and photosynthesis of rice. It causes a decrease in the rate of nutrient functioning and rice grain fullness.

In addition, the effects of control variables of fertilizer intensity, pesticide intensity, mechanical inputs, age, agricultural labor, rice scale, irrigation condition, and variety on rice yield also passed the significance test. Of these, the production input elements are undoubtedly the key to determining the level of yield. Rice yield is lower for older farmers and higher for households with more agricultural labor. The possible reason for this is that rice production stages such as transplanting, weeding, and draining still require a large amount of labor input, but older farmers generally have reduced physical strength and mobility, leading to a decrease in the fineness and effectiveness of paddy field management. Farmers with larger scale of rice production and more mechanization inputs have higher rice yield. Compared to traditional methods, scaled mechanized agricultural practices yield superior efficiency in rice production processes, and the standardized unified management mode will promote the increase of rice yield. Irrigation conditions, on the other hand, are the key to the control of water resources in the rice production process and an important factor in determining the quality of cultivated land. Differences in rice yield and quality can cause farmers to adopt different double or single cropping systems. However, from the survey, rice yield after two rounds of double-cropping are still generally higher than single-cropping harvests.

TABLE 2 Test results of pesticide reduction ratio threshold.

Threshold test	*F*-value	*p*-value	Boundary value			Threshold value	95% confidence interval
			1%	5%	10%		
Single-threshold	18.521**	0.032	57.282	38.194	20.561	33.250	(33.168, 33.342)
Double-threshold	5.447	0.210	53.347	22.584	16.227	50.637	(50.627, 50.646)

**Represents passing the 5% significance level test.

TABLE 3 Results of threshold regression estimation.

Variables	M1: Rice yield		M2: Rice yield	
	Coefficient	S.E.	Coefficient	S.E.
Pesticide reduction ratio ($reduce_i \leq 33.250$)	−1.327	0.989	−1.215	1.266
Pesticide reduction ratio ($reduce_i > 33.250$)	−2.073***	0.562	−1.976***	0.559
Drought	−3.732***	1.109	−3.511***	1.204
Flood	−1.226**	0.588	−1.172**	0.534
High temperature	−0.975***	0.235	−0.896**	0.384
Frost	−1.051**	0.457	−0.973*	0.523
Pesticide intensity			0.052*	0.028
Fertilizer intensity			0.197***	0.062
Mechanical input			1.622*	0.867
Gender			2.215	4.259
Age			−1.353**	0.621
Education			2.082	1.842
Family income			1.794	1.180
Agricultural labor			1.254*	0.709
Rice scale			0.479**	0.215
Irrigation condition			3.189***	1.003
Variety			2.276***	0.535
Henan			1.353	1.025
Constant	15.207***	3.059	5.691*	3.234
Chi-square	36.280***		27.760***	
Adjusted R^2	0.056		0.172	

*, **, and *** indicate that the estimated coefficients of the regression pass the 10, 5, and 1% significance level tests, respectively.

3.2 Robustness test

In the above empirical tests, the pesticide reduction ratio is used as the core indicator to examine its nonlinear impact on rice yield while controlling for pesticide intensity. Considering the diversity of ways to measure pesticide reduction indicators (Hou et al., 2020; Yan et al., 2024), this paper further verifies the stability of the empirical results by replacing the core variables. One is a threshold estimation using pesticide intensity as the core explanatory variable. The other is a threshold estimation using the absolute amount of pesticide reduction as the core variable. The results obtained are shown in Table 4. The threshold value for the effect of pesticide intensity on rice yield is 0.16 kg/mu. This indicates that moderate pesticide intensity is associated with increased rice yield, but the positive effect of pesticide intensity on rice yield diminishes beyond the threshold. Similarly, the threshold value for the effect of absolute pesticide reduction on rice yield is 0.09 kg/mu. From the segmentation results, the negative effect of excessive pesticide reduction on rice yield passed the 5% significance level test. The results are largely consistent with those in Table 3, verifying the robustness of the empirical results. Additionally, we supplemented another method for testing sample truncation robustness. Specifically, we removed elderly farmers (aged over 60) from the sample and re-estimated the results. This is because elderly respondents may exhibit greater memory bias when completing questionnaires, leading to increased measurement error in the indicators. The results of the robust estimates are presented in the last two columns of Table 4, which continue to support the aforementioned conclusions.

3.3 Heterogeneity impact of extreme climate

Considering that extreme climates can alter rice and pest survival to a greater extent, we included the interaction term between extreme climates and pesticide reduction ratios in the model. The results in Table 5 are obtained from regression estimation of the sample data using Equation 6. It showed that the interaction terms of drought, flood, and high temperature with pesticide reduction ratio on rice yield passed the significance level tests of 1, 10, and 10% respectively, and the effect was consistent with the pesticide reduction ratio indicator. This indicates that rice yield losses from pesticide reductions are further exacerbated under the three extreme climatic conditions of drought, flood, and high temperature, validating the research hypothesis H2. The risk of food yield losses from pesticide reductions under extreme climatic conditions is very high. Therefore, pesticide reduction measures during drought, flood, or high-temperature periods should be carefully evaluated compared to normal climatic conditions. On the one hand, drought, flood, and high temperatures can directly affect the normal growth of rice and

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TABLE 4 Model robustness test results.

Variables	M3: Rice yield		M4: Rice yield		M5: Rice yield	
	Coefficient	S.E.	Coefficient	S.E.	Coefficient	S.E.
Pesticide intensity ($pests_i \leq 0.16$)	2.150***	0.376				
Pesticide intensity ($pests_i > 0.16$)	1.073	0.762				
The absolute amount of pesticide reduction ($reduce_i \leq 0.09$)			−0.596	0.481		
The absolute amount of pesticide reduction ($reduce_i > 0.09$)			−0.287**	0.105		
Pesticide reduction ratio ($reduce_i \leq 33.250$)					−0.975	1.450
Pesticide reduction ratio ($reduce_i > 33.250$)					−0.713**	0.232
Drought	−0.142**	0.057	−2.072***	0.205	−0.459**	0.218
Flood	−0.632**	0.294	−0.455*	0.260	−0.514*	0.270
High temperature	−1.725***	0.536	−0.509**	0.217	−0.672***	−0.125
Frost	−0.271	0.203	−0.110	0.258	−0.490	0.871
Chi-square	28.354***		25.176***		32.306***	
Adjusted R^2	0.710		0.982		1.584	
Observations	1,120		1,120		865	

*, **, and *** indicate that the estimated coefficients of the regression passed the 10, 5, and 1% significance level tests, respectively. Other variables were controlled in the model and are not presented in this table.

TABLE 5 Interactive effects of extreme climate and pesticide reduction ratio on rice yield.

Variables	M6: Rice yield		M7: Rice yield		M8: Rice yield		M9: Rice yield	
	Coefficient	S.E.	Coefficient	S.E.	Coefficient	S.E.	Coefficient	S.E.
Pesticide reduction ratio	−2.154***	0.740	−1.786***	0.494	−2.031**	0.775	−1.924**	0.929
Drought	−2.817***	0.894						
Pesticide reduction ratio × Drought	−1.250***	0.415						
Flood			−0.557**	0.274				
Pesticide reduction ratio × Flood			−1.359*	0.727				
High temperatures					−0.627*	0.356		
Pesticide reduction ratio × High temperature					−0.867*	0.456		
Frost							−0.652*	0.377
Pesticide reduction ratio × Frost							−0.273	0.202
Other variables	Controlled		Controlled		Controlled		Controlled	
Chi-square	25.674***		21.527***		19.638***		20.871***	
Adjusted R^2	0.631		0.529		0.710		0.720	

*, ** and *** indicate that the estimated coefficients of the regression passed the 10, 5, and 1% significance level tests, respectively.

reduce its immunity against pests and diseases. On the other hand, extreme climate can upset the ecological balance by reducing the amount of natural enemies. This indirectly promotes the breeding of pests and causes an increased probability of rice yield loss.

The effect of frost and pesticide reduction ratio interaction terms on rice yield did not pass the significance test. This indicates that rice yield losses from pesticide reductions do not increase significantly under frost conditions. The above results may be due to the following two reasons: First, the sample sites of this study were selected mainly in central China. Both Hubei and Henan provinces have subtropical climates and do not experience severe low-temperature frosts. Severe frosty weather mainly occurs in the late autumn period. This mainly affects the tasseling and flowering period of rice growth, and the negative impact is extremely limited, so the impact on rice yield will not be too great. Second, frost is also unfavorable to some extent to the growth and breeding of insect pests, and may even suppress and kill some soil and plant pests and diseases. Sample farmers in the survey also generally reported that rice grown in warmer periods used more pesticide to manage pests than rice grown in cooler periods.

3.4 The mechanism of pesticide reduction effects on rice yield

Next, we test the hypothesis that pest outbreaks induced by pesticide reductions are a key driver of rice yield loss. Rice pest outbreaks are characterized by diversity and complexity, resulting in pest populations such as rice *planthopper*, *stem borer*, *magnaporthe grisea*, and *ustilaginoidea virens* which cannot be precisely quantified by farmers. In this study, the severity of pest infestation in rice—defined as the extent of crop damage reflected by both the spatial scale (area affected) and the temporal intensity (frequency of outbreaks)—was measured using the area and frequency of pest outbreaks, based on the study of Cheng et al. (2025). The questions in the questionnaire, "Percentage of the area with pest outbreaks during the rice growing cycle" and "Number of pest outbreaks during the rice growing cycle" were used to measure these two indicators. The estimation of the effect of pesticide reduction ratio on the area and frequency of pests under different sample setup scenario conditions such as full-sample, drought, flood, high temperature, and frost was executed using Stata software and the results are shown in Table 6.

In the full-sample estimation, the effects of moderate pesticide reductions on both the area and frequency of pests did not pass the significance test. However, the effects of excessive pesticide reductions above the threshold value on the area and frequency of pests passed significance tests, respectively. It is indicated that pesticide reductions exceeding the limit values induce an increase in the area and frequency of pests. This also indicates that excessive pesticide reduction leads to rice yield loss. From the results estimated under the drought scenario, the effect of excessive pesticide reduction on both the area and frequency of pests passed significance test, and the effect of moderate pesticide reduction on both the area and frequency of pests passed significance test. Unlike the full-sample estimates, pesticide reduction leads to pest outbreaks. Similar results are presented in the flooding scenario. Pesticide reductions significantly increase the probability of pest outbreaks under drought and flood conditions. The sample farmers in the survey also pointed out that their production experience is that they would choose to use more pesticide to secure grain yield in drought and flood climates. The effects of excessive pesticide reductions on both the area and frequency of pests under high temperature scenarios passed the 10% significance test, while the effects of moderate pesticide reductions were not significant. It indicated that pesticide reduction under high temperature also increased the risk of rice pest outbreaks. Meanwhile, the effect of pesticide reduction under frost scenario on the area and frequency of pests was not significant. Table 6 is harmonized with the results in Table 5. These results can reasonably explain that drought, flood, and high temperature climates can induce large and high frequency outbreaks of pests, which in turn lead to an increased probability of rice yield loss due to pesticide reductions. That is, drought, flood, and high temperature can create greater impediments to pesticide reductions under food security objectives.

4 Discussion

This study found that excessive pesticide reduction increases risks to food security stability. The goal of "zero growth" in pesticide use in China has now been realized. Whether the Chinese government needs to continue mandating the "negative growth" target for pesticide use. The key to answering these two questions is whether China's pesticide reduction ratio is at the threshold, i.e., whether it is below the scientific dose for effective pest management. Our research team concluded that the answer is no. That is, China should still insist on setting scientific goals for continued pesticide reductions (Cai et al., 2021; Zhang et al., 2022; Shao et al., 2024).

On the one hand, China's pesticide reduction ratio from 2015 to 2020 is 1.13, 2.38, 4.91, 9.15, 7.44, and 5.63%, respectively. The cumulative reduction ratio for these 5 years has reached 27.32%. There is no doubt that pesticide reduction in China has achieved great achievements. However, FAO data indicate that China's pesticide intensity in 2020 still is 13.07 kg/ha, much higher than the US's 2.54 kg/ha, the UK's 3.17 kg/ha, and France's 4.45 kg/ha. That is, although the total growth of pesticide in China has been controlled, the intensity of pesticide is still very high. At the same time, China's

TABLE 6 Effect of pesticide reduction on the outbreak and frequency of rice pest.

Observation	Variables	M10: Pest area		M11: Pest frequency	
		Coefficient	S.E.	Coefficient	S.E.
Full-sample	Pesticide reduction ratio ($reduce_i \leq \gamma$)	0.526	0.4208	0.151	0.274
	Pesticide reduction ratio ($reduce_i > \gamma$)	0.257**	0.125	0.107***	0.036
Drought	Pesticide reduction ratio ($reduce_i \leq \gamma$)	0.367**	0.147	0.258**	0.106
	Pesticide reduction ratio ($reduce_i > \gamma$)	0.294***	0.054	0.163***	0.052
Flood	Pesticide reduction ratio ($reduce_i \leq \gamma$)	0.351**	0.168	0.196**	0.078
	Pesticide reduction ratio ($reduce_i > \gamma$)	0.407***	0.129	0.104***	0.032
High temperature	Pesticide reduction ratio ($reduce_i \leq \gamma$)	0.259	0.462	0.160	0.340
	Pesticide reduction ratio ($reduce_i > \gamma$)	0.317*	0.188	0.078*	0.042
Frost	Pesticide reduction ratio ($reduce_i \leq \gamma$)	0.385	0.433	0.128	0.096
	Pesticide reduction ratio ($reduce_i > \gamma$)	0.296	0.228	0.099	0.065

*, **, and *** indicate that the estimated coefficients of the regression passed the 10, 5, and 1% significance level tests, respectively.

three major grain crops' pesticide use efficiency is only an average of 39.8%, far lower than the developed countries of 50–60%. Therefore, pesticide reduction in China needs to be sustained.

On the other hand, this study utilized macro data for further statistics on pesticide reduction and rice yield in Hubei and Henan Provinces (Figure 3). At present, pesticide reduction actions are continuing in the above two provinces, but rice yields have not shown a clear downward trend. Obviously, the pesticide reduction ratio still has not reached the real threshold value, and the potential for pesticide reduction is still very large. However, it is undeniable that the phenomenon of rice yield loss due to excessive pesticide reduction was observed at the micro level in this study. This is mainly due to the fact that rice is the ration crop, and sample farmers are heavily reducing pesticide or even not using pesticide to have lower pesticide residues in their rations. Because food safety is more important than yield for them, especially for small-scale farmers (Qin and Lv, 2020). Certainly, pesticides remain essential inputs in the production of commercial grains today. The findings of this study are good to provide practical references for the future sustainable pesticide reduction in China. These findings serve as a warning for pesticide reductions under food security, thus avoiding blindly aggressive pesticide reductions in various regions.

China's ongoing efforts to reduce pesticide use are strongly supported by international precedents. For instance, the European Union has implemented its "Farm to Fork" strategy, which sets a binding target to reduce overall chemical pesticide use by 50% by 2030. This initiative stems from recognition of the environmental and health costs associated with high-intensity pesticide reliance, echoing China's own sustainable development challenges. Similarly, within Asia, the widespread adoption of Integrated Pest Management (IPM) offers a viable pathway for pesticide reduction. For instance, Indonesia's national rice production program has successfully reduced insecticide use by over 50% while maintaining or even increasing yields through fostering ecological balance and utilizing natural enemies. These international examples demonstrate that combining mandatory reduction targets with scientific pest management approaches is not merely theoretical—it is actively being implemented in both developed and developing countries. They validate the correct direction of China's policy.

The discussion of these findings raises two points for consideration: First, it is recommended that farmers in non-extreme climate years attempt to reduce pesticide application to the "safe threshold" range identified in this study. Integrated Pest Management (IPM) strategies—such as planting resistant varieties, adopting conservation tillage to enhance crop resilience, and introducing natural enemies—should be prioritized as alternatives or supplements to chemical pesticides, rather than simply reducing all pesticide use proportionally. Second, policymakers should move beyond simplistic, one-size-fits-all reduction targets and instead support more flexible, climate-condition-based pesticide reduction strategies. Specifically, establish regional early warning systems for extreme weather and pest outbreaks, providing farmers with targeted, temporary application guidance upon alerts; link agricultural insurance to the adoption of green pest control practices; and increase subsidies for ecological agriculture and agronomic innovation to fundamentally reduce reliance on chemical pesticides.

5 Limitations

It should be noted that the subjects of this study are rice growers in China, and the sample is mainly smallholder farmers. The particularity of the samples is as follows: Firstly, the research samples are distributed across China, a nation currently undergoing mandatory pesticide reduction. This distinctly differentiates it from other traditional agricultural countries. The limitation of our sample lies in selecting only Hubei and Henan provinces within China, which may somewhat affect the generalizability of our findings. However, as representative regions of China's major rice-producing areas—the middle and lower reaches of the Yangtze River and North China respectively—the issues examined hold high relevance within these specific contexts. The mechanistic conclusions and methodological

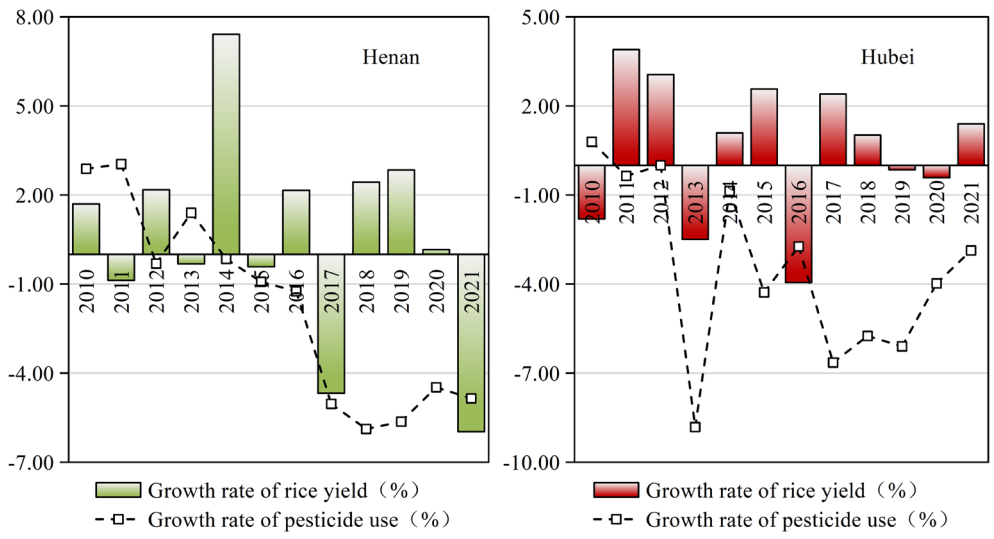


FIGURE 3
Pesticide reduction and rice yield growth rate in Hubei and Henan province.

framework derived from this study can provide important theoretical and practical references for conducting similar research in other comparable agricultural regions. Secondly, the sample primarily consists of smallholder farmers, which implies that the findings may not be entirely generalizable to larger-scale commercial grain producers such as family farms, cooperatives, and agricultural enterprises. Thirdly, these smallholder farmers predominantly cultivate rice as a staple crop, indicating that their focus is less on maximizing yield and more on ensuring food safety. Consequently, there is a prevalent tendency to reduce pesticide usage, a decision that would be deemed impractical for commercial grain growers. Fourth, some indicators in the regression model rely on farmer self-reports, which may introduce recall bias. Examples include the indicators for pest outbreak area and frequency. Data limitations have led the study to focus more on analyzing observable pest outbreaks that reach economic thresholds, while falling short in providing a more detailed portrayal of species diversity dynamics. Future research should incorporate objective methods such as field monitoring data or remote sensing data to more accurately assess the severity of pest damage. Similarly, this study defined climate variables using dummy variables, which only provides a preliminary exploration of their moderating effects. Incorporating scientifically instrumented climate characteristics such as intensity, duration, and timing data in future research would further enhance the scientific value of the findings. Of course, these limitations do not negate the value of the findings in this study. This study provides a warning for the cultivation of commodity grains. This also avoids blind pesticide reduction by local governments, which increases food security risks. Additionally, it should be noted that this study does not attempt to comprehensively measure other dimensions of food security, such as farmer income, market access, or nutritional health. These important aspects warrant further exploration in future research.

6 Conclusion

Based on survey data from the main rice producing areas in Hubei and Henan provinces of China, this study examined the impact of pesticide reductions on rice yield using a threshold effect model. The dynamics of this effect were also explored under extreme climate scenarios such as drought, flood, high temperature, and frost. The main conclusions obtained are as follows:

There was a single threshold effect of the pesticide reduction ratio of the sample rice farmers on rice yield, with a threshold value of 33.25% for the pesticide reduction ratio. This indicates that there is no significant correlation between moderate reductions in pesticide use and decreases in rice yields, but excessive reductions in pesticide use may carry a higher risk of rice yield losses. In addition, age, agricultural labor, rice scale, mechanical input, irrigation condition, and rice variety are also important factors affecting rice yield.

There was a significant interaction effect between extreme climate (drought, flood, high temperature, and frost) and pesticide reduction on rice yield. This suggests that the occurrence of extreme climates can further intensify rice yield losses due to pesticide reduction. In particular, drought, flood, and high temperature climates can induce larger area and more frequent outbreaks of pests, which puts rice yields at greater risk of loss. These findings provide preliminary evidence that pesticide reduction may face challenges under extreme weather conditions.

In light of the demonstrated findings, strategic policy recommendations are formulated as follows: First, local governments should formulate scientific and reasonable medium- and long-term pesticide reduction plans to avoid blindly venturing into pesticide reduction. It is necessary to insist on the use of scientific and rational ways, such as technical training, application guidance, experimental demonstration, and other agronomic extension activities, to regulate the application of pesticide by farmers. This can reduce the use of pesticides over the standard and is beneficial to the continued promotion of pesticide reduction in China. At the same time, it is recommended that a small-scale demonstration of pesticide reduction be carried out in the main food-producing regions first, and then gradually expand the effectiveness of pesticide reduction under the circumstance of guaranteeing the stability of grain yield. It is recommended that policymakers should develop a pesticide reduction program of appropriate scope and extent in an appropriate period to prevent some regions from making quick successes and blindly carrying out drastic pesticide reduction actions. Second, it is necessary to set up dynamic optimization and adjustment strategies for each region to cope with the impact of extreme climate change on pesticide reduction. The country's geographic differences and the uneven distribution of water resources, have led to frequent drought, flood, and high temperature. These pose great challenges to food security stability and also hinder pesticide reduction efforts. It is recommended that policymakers adopt a medium- to long-term pesticide reduction target assessment system. In the short term, they should increase the flexibility of the pesticide reduction target assessment of local governments, to make the pesticide reduction work more flexible and the risk more controllable.

Data availability statement

The datasets presented in this study can be found in online repositories. The names of the repository/repositories and accession number(s) can be found in the article/supplementary material.

Ethics statement

The studies involving humans were approved by the Academic Ethics Committee of the School of Law and Business, Wuhan University of Technology. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

YH: Conceptualization, Funding acquisition, Methodology, Software, Writing – original draft. LT: Data curation, Investigation, Writing – review & editing. YS: Data curation, Formal analysis, Methodology, Visualization, Writing – original draft.

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Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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